

The General Theory of Signal Extraction

A Unified Bayesian, Physical, Statistical, Informational, and Instrumentation Framework

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Abstract

Signal extraction is one of the most fundamental problems in science, engineering, economics, finance, warfare, intelligence gathering, medicine, psychology, and artificial intelligence.

A measurement device receives a noisy observation generated by an underlying information-bearing process and an instrument-dependent carrier mechanism.

This paper develops a unified mathematical theory of signal extraction based upon operator theory, Bayesian inference, information theory, state-space systems, inverse problems, geometry, and statistical decision theory.

The resulting framework encompasses classical communications, instrumentation theory, Kalman filtering, Wiener filtering, machine learning, sensor fusion, economic signal extraction, battlefield intelligence systems, biological sensing, and scientific measurement as special cases.

The paper ends with “The End”

Contents

1	Introduction	1
2	The General Observation Equation	2
2.1	Interpretation	2
3	Examples	2
3.1	Amplitude Modulation	2
3.2	Phase Modulation	3
3.3	Frequency Modulation	3
3.4	General Nonlinear Modulation	3
4	White Noise	3
4.1	Physical Interpretation	3
5	Bayesian Signal Extraction	3
5.1	Optimal Estimator	4
6	Hilbert Space Theory	4
7	Spectral Theory	4
7.1	Wiener Filter	4

8 Information Theory	5
8.1 Perfect Recovery	5
8.2 Complete Failure	5
9 Dynamic Signal Extraction	5
9.1 Filtering Objective	5
10 Kalman Filtering	5
11 Geometric Signal Extraction	5
11.1 Projection Map	6
12 Statistical Learning Interpretation	6
13 Applications Across Scientific Domains	6
13.1 Physics	6
13.2 Chemistry	6
13.3 Biology	7
13.4 Economics	7
13.5 Finance	7
13.6 Psychology	7
13.7 Psychiatry	8
13.8 Political Science	8
13.9 International Relations	8
13.10 Warfare Economics	8
13.11 Military Operations	8
14 Computational Framework	9
14.1 General Optimization Problem	9
15 Pseudo-Code	9
16 A Universal Signal Extraction Principle	9
17 Illustrative Architecture Diagram	10
18 Comparative Domain Table	10
19 Conclusion	10

List of Figures

1 General signal extraction architecture.	10
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List of Tables

1 Signal extraction across disciplines.	10
---	----

1 Introduction

Signal extraction lies at the heart of nearly every empirical discipline.

Examples include:

- Radio communication.
- Radar systems.
- Sonar systems.
- Medical diagnostics.
- Financial forecasting.
- Macroeconomic measurement.
- Military intelligence.
- Neuroscience.
- Climate science.
- Artificial intelligence.

The common objective is simple:

Recover the informative signal from noisy observations.

This paper develops the most general mathematical framework presently known for this objective.

2 The General Observation Equation

Definition 1 (General Instrument Operator). *Let*

$$S(t)$$

denote an information-bearing process,

$$\Theta(t)$$

*denote the carrier or instrument state,
and*

$$N(t)$$

denote white noise.

The most general observation equation is

$$Y(t) = \mathcal{I}(S(t), \Theta(t)) + N(t),$$

where

$$\mathcal{I}$$

is an arbitrary instrument operator.

2.1 Interpretation

The framework separates measurement into three components:

1. Information.
2. Carrier mechanism.
3. Noise.

This decomposition applies to nearly all known sensing systems.

3 Examples

3.1 Amplitude Modulation

$$Y(t) = A(t)S(t) + N(t).$$

3.2 Phase Modulation

$$Y(t) = S(t) \cos(\omega t + \phi) + N(t).$$

3.3 Frequency Modulation

$$Y(t) = A \cos \left(\omega_c t + k \int_0^t S(\tau) d\tau \right) + N(t).$$

3.4 General Nonlinear Modulation

$$Y(t) = \mathcal{I}(S, \Theta) + N.$$

4 White Noise

Definition 2. *White noise is a stochastic process satisfying*

$$E[N(t)] = 0$$

and

$$E[N(t)N(s)] = \sigma^2 \delta(t - s).$$

4.1 Physical Interpretation

White noise emerges from:

- Thermal fluctuations.
- Quantum effects.
- Molecular collisions.
- Electronic noise.
- Human cognitive uncertainty.

5 Bayesian Signal Extraction

The objective is to estimate

$$P(S|Y).$$

By Bayes' theorem,

$$P(S|Y) = \frac{P(Y|S)P(S)}{P(Y)}.$$

5.1 Optimal Estimator

Under quadratic loss,

$$\hat{S} = E[S|Y].$$

Proposition 1. *The posterior mean minimizes expected squared error.*

Proof. Consider

$$L = E[(S - \hat{S})^2].$$

Differentiating with respect to $\hat{S}$ gives

$$\hat{S} = E[S|Y].$$

□

6 Hilbert Space Theory

Let

$$\mathcal{H} = L^2([0, T]).$$

Then

$$Y = S_C + N.$$

Signal extraction becomes projection.

Theorem 1. *The optimal linear estimator is the orthogonal projection*

$$\hat{S} = P_S(Y).$$

Proof. Orthogonal projection minimizes norm distance in Hilbert spaces.

□

7 Spectral Theory

Taking Fourier transforms,

$$Y(\omega) = S_C(\omega) + N(\omega).$$

7.1 Wiener Filter

$$H_W(\omega) = \frac{S_S(\omega)}{S_S(\omega) + S_N(\omega)}.$$

Theorem 2. *The Wiener filter minimizes mean-square reconstruction error among all linear filters.*

Proof. Direct minimization of expected quadratic error yields the result. □

8 Information Theory

Define mutual information

$$I(S; Y).$$

8.1 Perfect Recovery

Perfect extraction occurs when

$$I(S; Y) = H(S).$$

8.2 Complete Failure

Complete failure occurs when

$$I(S; Y) = 0.$$

9 Dynamic Signal Extraction

Consider

$$dX_t = f(X_t, t)dt + Q^{1/2}dW_t$$

and

$$dY_t = h(X_t, t)dt + R^{1/2}dV_t.$$

9.1 Filtering Objective

Estimate

$$P(X_t|Y_t).$$

10 Kalman Filtering

For linear systems,

$$X_{t+1} = AX_t + \varepsilon_t$$

and

$$Y_t = HX_t + \eta_t.$$

Kalman filtering provides recursive optimal estimation.

11 Geometric Signal Extraction

Define the carrier manifold

$$\mathcal{M} = \{\mathcal{I}(S, \Theta)\}.$$

Observed data are noisy points on $\mathcal{M}$.

11.1 Projection Map

$$\pi : \mathcal{M} \rightarrow \mathcal{S}.$$

Signal extraction becomes coordinate recovery.

12 Statistical Learning Interpretation

Suppose

$$Y = f_{\theta}(S) + N.$$

Machine learning seeks

$$\hat{f}^{-1}.$$

Examples include:

- Neural networks.
- Transformers.
- State-space models.
- Bayesian networks.
- Diffusion models.

13 Applications Across Scientific Domains

13.1 Physics

Examples:

- Radar.
- Quantum measurement.
- Gravitational wave detection.
- Radio astronomy.

13.2 Chemistry

Examples:

- Spectroscopy.
- NMR.
- Mass spectrometry.
- Chemical sensing.

13.3 Biology

Examples:

- Genetic sequencing.
- Neural coding.
- Cellular signaling.
- Sensory systems.

13.4 Economics

Observed variables satisfy

$$Y_t = Y_t^* + \epsilon_t.$$

where

$$Y_t^*$$

is latent economic activity.

Examples:

- GDP estimation.
- Inflation measurement.
- Productivity estimation.
- Business-cycle extraction.

13.5 Finance

Examples:

- Price discovery.
- Latent volatility estimation.
- Market microstructure.
- Order-flow inference.

13.6 Psychology

Human perception performs signal extraction continuously.

Examples:

- Visual perception.
- Auditory processing.
- Decision making.
- Attention mechanisms.

13.7 Psychiatry

Many psychiatric disorders may be interpreted as distortions of internal signal extraction processes.

Examples:

- Hallucinations.
- Delusions.
- Sensory misclassification.

13.8 Political Science

Governments infer:

- Public sentiment.
- Electoral trends.
- Social stability.

from noisy data.

13.9 International Relations

States attempt extraction of:

- Strategic intent.
- Military readiness.
- Economic strength.
- Diplomatic signalling.

13.10 Warfare Economics

Signal extraction is fundamental to:

- Resource allocation.
- Logistics estimation.
- Strategic forecasting.
- Battlefield coordination.

13.11 Military Operations

Examples:

- Radar.
- Sonar.
- Electronic warfare.
- Signals intelligence.
- Satellite reconnaissance.

14 Computational Framework

14.1 General Optimization Problem

$$(\hat{S}, \hat{\Theta}) = \arg \min [-\log P(Y|S, \Theta) - \log P(S, \Theta)].$$

15 Pseudo-Code

Input:

Observed data Y

Initialize:

Signal estimate S

Carrier estimate Θ

Repeat:

Compute likelihood

Update posterior

Update signal estimate

Update carrier estimate

Until convergence

Output:

Recovered signal

16 A Universal Signal Extraction Principle

Theorem 3 (Universal Signal Extraction). *Every measurement system may be represented as*

$$Y = \mathcal{I}(S, \Theta) + N.$$

Signal extraction consists of recovering

S

from

Y .

Proof. All measurement systems contain:

1. Information source.
2. Carrier mechanism.
3. Noise process.
4. Observation channel.

Hence all may be represented by the stated equation.

□

17 Illustrative Architecture Diagram

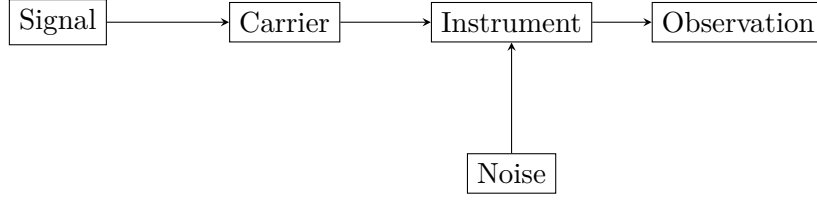


Figure 1: General signal extraction architecture.

18 Comparative Domain Table

Domain	Signal	Noise
Physics	Waveform	Thermal Noise
Chemistry	Spectrum	Instrument Noise
Biology	Neural Activity	Biological Variability
Economics	Latent Output	Measurement Error
Finance	Fundamental Value	Market Noise
Psychology	Sensory Input	Cognitive Noise
Military	Target Position	Countermeasures
AI	Latent Feature	Data Noise

Table 1: Signal extraction across disciplines.

19 Conclusion

Signal extraction constitutes a universal scientific problem. The framework developed herein unifies communications theory, measurement science, Bayesian inference, machine learning, economic measurement, intelligence gathering, and scientific instrumentation within a single mathematical architecture.

The central equation

$$Y = \mathcal{I}(S, \Theta) + N$$

provides a general representation of all measurement systems, while Bayesian inversion provides the universal extraction mechanism.

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Glossary

Signal Information-bearing process.

Carrier Mechanism transporting information.

Noise Random disturbance contaminating observations.

Instrument Operator Mapping from signal space to observation space.

Posterior Conditional probability after observing data.

Mutual Information Information shared by two variables.

Wiener Filter Optimal linear filter.

Kalman Filter Optimal recursive estimator for linear Gaussian systems.

Carrier Manifold Geometric representation of instrument states.

The End